

Community Forensics: Using Thousands of Generators to Train Fake Image Detectors

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Abstract

One of the key challenges of detecting AI-generated images is spotting images that have been created by previously unseen generative models. We argue that the limited diversity of the training data is a major obstacle to addressing this problem, and we propose a new dataset that is significantly larger and more diverse than prior works. As part of creating this dataset, we systematically download thousands of text-to-image latent diffusion models and sample images from them. We also collect images from dozens of popular open source and commercial models. The resulting dataset contains 2.7M images that have been sampled from 4803 different models. These images collectively capture a wide range of scene content, generator architectures, and image processing settings. Using this dataset, we study the generalization abilities of fake image detectors. Our experiments suggest that detection performance improves as the number of models in the training set increases, even when these models have similar architectures. We also find that increasing the diversity of the models improves detection performance, and that our trained detectors generalize better than those trained on other datasets. The dataset can be found in

https://jespark.net/projects/2024/community_forensics

1. Introduction

Our ability to automatically generate realistic images is quickly outpacing our ability to detect them, potentially leading to a state of affairs in which neither humans nor machines can reliably tell real from fake. While the field of image forensics has been developing methods to address this problem, existing fake image detectors still struggle with generalization. These methods often excel at detecting images from generators that were present in their training sets, but fail when given images sampled from unseen models [87, 101, 102, 125].

A core challenge is dealing with the large amounts of variation between models. Each generator has a potentially unique combination of the architecture, loss function, and training distribution. Even seemingly minor differences in

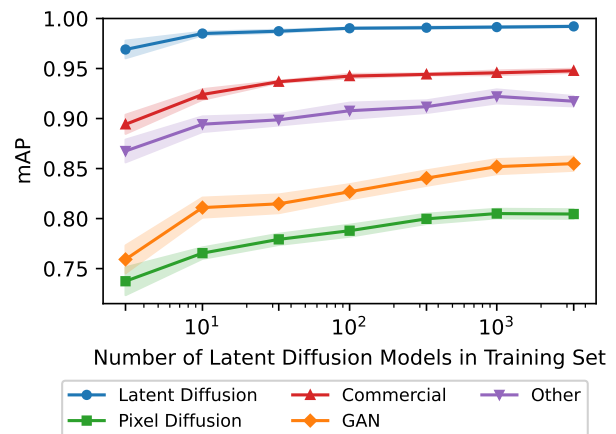


Figure 1. **Performance vs. model diversity.** We use images sampled from different numbers of open source latent diffusion models in our *Community Forensics* dataset to train fake image detectors (shown in Fig. 2a). We fix the total number of images and only vary the number of models. The detector’s performance increases across all generator types as we train from more models, even though these added models are entirely latent diffusion. This improvement is largest for test images from out-of-distribution generators, such as pixel-based diffusion models or GANs.

low-level image processing details, such as the ways that training images are resized or compressed, can strongly influence detection accuracy [125]. As a result of these model-specific idiosyncrasies, a generator’s images may evade detection, even when images from architecturally similar models exist in the training set. This issue has been exacerbated by the thousands of open source models that are now available online, many of which extend pretrained base models in complex ways.

We hypothesize that the lack of diversity in training datasets is a major source of these shortcomings. Although today’s datasets often contain millions of fake images, they come from a relatively small number of generators. As a result, this data fails to capture many sources of variation that one might encounter in the wild. These limitations also make it challenging to accurately benchmark performance, since it is easy for cues that work well on one set of generators to fail on others.



Figure 2. **The Community Forensics dataset.** Our dataset contains images sampled from three types of generative models. (a) We systematically download open-source latent diffusion models from a model-sharing community [34, 123]. (b) We select popular open source generators with a variety of architectures and training procedures. (c) We sample from both closed and open state-of-the-art commercial models. We present example images and their corresponding model names.

To address these problems, we propose *Community Forensics*, a dataset that is significantly more diverse and comprehensive than those in prior works (Fig. 2). Our dataset contains images generated by: (a) thousands of systematically downloaded open-source latent diffusion models, (b) hand-selected open source models with various architectures, and (c) state-of-the-art commercial models.

To acquire large numbers of models, we sample images from thousands of text-to-image diffusion models hosted on a popular model-sharing website, Hugging Face [34]. We exploit the fact that these models use a common programming library [123] and thus can be sampled in a standardized way. A large fraction of them are extensions of Stable Diffusion [100], but collectively capture a variety of common model variations, such as in the architecture, image processing, and image content. We also sample images from many other open source models, including GANs [37], autoregressive models [39], and consistency models [79, 114]. To ensure sufficient diversity in the image content of the generated images, we sample generators using captions sourced from a mix of several existing datasets whenever possible.

Our dataset contains 4803 distinct models, roughly $34\times$ more than the most extensive previous dataset that sample images from generative models [5, 7, 15, 32, 43, 87, 125, 133], and covers a variety of recent model designs (Fig. 2).

We use this dataset to study generalization in the generated image detection problem. Our experiments confirm

that increasing the diversity of data improves generalization, consistent with previous findings in other areas of computer vision [48, 107, 108]. In contrast to other work, our dataset aims to increase diversity via *number of generators*, an underexplored axis of diversity. Through experiments, we find:

- *Classifiers trained on our dataset obtain strong performance*, both on our newly proposed evaluations and on multiple previously-proposed benchmarks.
- *Adding more generative models improves generalization.* Fig. 1 demonstrates the performance of fake image detection when trained on samples from varying numbers of diffusion models. Notably, the performance improves as more models are added, even across different architectures.
- *Including diverse generative model architectures significantly improves results*, since classifiers do not fully generalize between generator architectures.
- *Standard classifiers perform well.* In contrast to observations from recent work, we find that end-to-end training of classifiers based on CNNs or ViTs generalizes well, with qualitatively similar to that of other recognition problems.

2. Related work

Datasets for detecting generated images. A number of datasets have been proposed for specifically detecting “deepfake” images containing manipulated faces [25, 63, 67, 71, 101, 102, 134]. Rather than focusing on face manipulation, we address creating general-purpose meth-

ods that can detect images that have been directly produced by generative models. Wang *et al.* [125] proposed a widely-used dataset of CNN-generated images, mixing images from GANs [8, 12, 58, 59, 90, 132] with other models [9, 10, 18, 70, 102]. This work showed that forensics models generalize between generative models, providing motivation for training on large datasets of diverse generators. However, their classifier was trained on images from a single GAN and was highly sensitive to data augmentation parameters, and more recent work shows that it does not generalize to newer models [14, 87]. Ojha *et al.* [87] introduced a dataset of recent diffusion models and found that training a linear classifier on CLIP features [96] extracted from ProGAN-generated images performed well. Cozzolino *et al.* [15] extend this work by studying the performance of CLIP-based detectors on various generative models and datasets. Epstein *et al.* [32] simulated detecting fake images in an online way by training a detector up to a certain year and testing it on generators released after that year. Zhu *et al.* [133] collected 1.4M generated images from 8 different generators. These datasets, however, only consider a handful of models (less than 20 each), limiting the generalization of their detectors. Asnani *et al.* proposed RED116 [5], collecting 116K images from 116 generative models for predicting the model hyperparameters of a given generated image. Guo *et al.* [43] extends RED116 to 140 generative models with 140K images total. We improve upon these works by collecting much more diverse generative models to improve the performance and generalization of the detector. In concurrent work, Hong *et al.* [49] acquires user-created images from Midjourney and CivitAI. This strategy is complementary to ours: while it aims to collect in-the-wild fake images, its distribution is centered on images that users share, and the models are not necessarily identifiable, making it challenging to rigorously analyze the dataset’s contents and to interpret experiments conducted on it.

Fingerprint-based image forensics methods. Classic work on image forensics relied on methods based on image statistics [95] and physical constraints [56], rather than learning. A number of datasets have been created for detecting images that have been manipulated using traditional methods, such as with photo editors [21, 26, 53, 64, 84]. Recent works focus on detecting synthetic images by inspecting the generator fingerprints. Zhang *et al.* [131] and Marra *et al.* [80] proposed identifying the spatial fingerprints left by the generator to detect synthetic images. Others focus on spectral anomalies to detect synthetic images. Durall *et al.* [29] and Dzanic *et al.* [30] identified that CNN-generated images fail to reproduce certain spectral properties of real images. Corvi *et al.* [14] study the frequency fingerprints of the generated images and analyzes the cross-architecture generalization of the detector. Bammey [7] uses high-frequency artifacts to detect generated images. However, these approaches may be

brittle since the artifacts they rely on can be eliminated by post-processing [15]. We instead approach this problem in a data-driven manner, scaling the number of models, images, and architectures. Recent work has created ensembles of fake image classifiers [50]. In parallel, researchers have detected text generated by language models using supervised learning and heuristics [6, 35, 55, 65, 82, 103, 113, 121], which closely resemble those in visual forensics. However, no existing techniques that we are aware of aim to collect comprehensive datasets of community-created generators.

Out-of-distribution generalization. Our work is related to the out-of-distribution recognition problem as it involves generalizing to unseen generators and image processing pipelines. A variety of approaches have been proposed for this problem, based on likelihood ratios [72, 98, 128], self-supervision [47, 83, 109, 124], internal model statistics [46, 104], temperature scaling [3, 73], and via energy-based models [28, 31, 75]. Works by Schuhmann *et al.* [108] and Hendrycks *et al.* [48] show that diverse training data and data augmentation are important to improving the robustness to out-of-distribution samples. Our results are in line with these conclusions, as we find that a diverse set of generative models and stronger augmentations improve generalization.

3. The Community Forensics Dataset

To support our goal of studying generalization in generated image detection, we collect a dataset of images sampled from a wide range of models (Fig. 2). Our dataset consists of: (a) a large and systematically collected set of “in-the-wild” text-to-image latent diffusion models obtained from a model-sharing website, (b) hand-selected models from other open source architectures, and (c) closed and open state-of-the-art commercial models. We also pair these generated images with real images from other datasets. For all images in our dataset, we preserve the original image format where possible, without any additional compression or re-sizing. This is to mitigate potential bias and performance degradation in out-of-distribution settings due to unwanted artifacts [40, 42, 97]. Our dataset contains significantly more models than previous works (Tab. 1) and spans a wider range of architectures, processing pipelines, and semantic content.

3.1. Systematically collecting generative models

We perform our systematic collection using publicly available, open source¹ models that use the Hugging Face *diffusers* library [34, 123] because: 1) it is a popular library for creating text-to-image models and is widely used by hobbyists, 2) thousands of such models are publicly indexed, and 3) it provides a standard interface by which we can sample images. We process the models in the order of popularity, as indicated by the number of downloads. Our

¹We use “open source” to describe models with public weights and code, even if they may be closed in some aspect (*e.g.*, private training data).

Dataset	Models	Images	Architectures	Training setup
Wang <i>et al.</i> [125]	11	362K	GAN, Perceptual, Deepfake, ...	ProGAN [58] vs. LSUN [129]
Ojha <i>et al.</i> [87]	4*	10K*	GAN, Perceptual, Diffusion, ...	ProGAN [58] vs. LSUN [129]
Epstein <i>et al.</i> [32]	14	570K	Diffusion	Diffusion vs. LAION [108]
Cozzolino <i>et al.</i> [15]	18	26K	Diffusion	LDM [100] vs. MS-COCO [74]
Synthbuster [7]	9	10K	Diffusion	Diffusion vs. Dresden [36]
GenImage [133]	8	1.4M	Diffusion, GAN	Diffusion, GAN vs. ImageNet [23]
RED116 [5]	116	116K	GAN, VAE, Autoregressive, ...	Many vs. Many
RED140 [43]	140	140K	Diffusion, GAN, VAE, ...	Many vs. Many
Ours	4803	2.7M	Diffusion, GAN, Autoregressive, ...	Many vs. Many

Table 1. **Comparison with existing forensics datasets.** We compare the size of the dataset with existing datasets containing identifiable generative models. We only count the number of generated images. Our dataset contains significantly more generative models than prior works. *: Only counting the unique evaluation set by Ojha *et al.* [87] as their dataset is based on Wang *et al.* [125].

pipeline downloads each model and extracts relevant hyperparameters (e.g., number of diffusion steps and guidance scale), sampling pipeline configurations, and metadata, from both the model cards on the model-sharing webpage and the metadata retrieved by the `diffusers` library [34, 123]. We sample images using a distribution of text prompts obtained from various real datasets (Sec. 3.3). Since experiments suggest diminishing returns for repeatedly sampling from any given model, we sample a few hundred images from each one. We sample 4763 models with approximately 403 images each, for a total of 1.9M images from this process.

While the lack of documentation in each model and the scale of data collection make it challenging to exactly characterize the model designs in this set, they appear to be entirely (or almost entirely) based on latent diffusion. More specifically, we categorize models as being based on *latent diffusion* if they perform a denoising process on a latent representation.² Based on this criterion and the self-reported tags, all models in our systematically collected set appear to be based on latent diffusion. While pixel-based diffusion models also use the `diffusers` library (e.g., DeepFloyd [22]), they were incompatible with our automated generation pipeline. We record such incompatible models and manually sample them to either construct an out-of-distribution test set (Sec. 3.4), or as manually-chosen models for training data (Sec. 3.2).

We show examples of the sampled images in Fig. 2. In the supplement, we provide examples of models and information from their project pages. These models generate a variety of images, with various types of semantic content and preprocessing. For example, a large fraction of these models adapt variations of a popular pretrained latent diffusion model, Stable Diffusion [100], to different downstream applications, and use a number of adaptation strategies (e.g., using LoRA [51]). We provide the model metadata with each image to enable other possible forensics applications. We discuss these in the supplement and provide information about image and model licenses.

²We note that this definition includes latent consistency models [78].

3.2. Collecting images from other architectures

Images from manually chosen models. To ensure that our dataset contains a broader range of models, we manually select 19 models from public repositories and sample an average of 40,738 images per model, resulting in 774K images total. We note that this number is itself on par with (or more than) many prior datasets with identifiable generative models [7, 15, 32, 87, 125, 133]. We include several GANs (e.g., StyleGANs [60–62, 106], BigGAN [8], StyleSwin [130], GigaGAN [57], ProGAN [58], ProjectedGAN [105], GANs-former [52], SAN [115], and CIPS [4]), pixel-based diffusion models (e.g., GLIDE [85], ADM [24], and DeepFloyd [22]), latent diffusion models (e.g., VQ-Diffusion [41], Diffusion Transformers [92], and Latent Flow Matching [20]), and an autoregressive model (Taming Transformers [33]).

Images from commercial models. We sample 15K images from 11 commercial models using LAION-based captions to evaluate the generalization to state-of-the-art models with typically unknown architectures: DALL-E 2, 3 [88, 89], Ideogram V1, V2 [2], Midjourney V5, V6 [81], Firefly Image 2, 3 [1], FLUX.1-dev, schnell [68], and Imagen 3 [38].

3.3. Collecting real images

To help study how real images influence forensics models, we source real images from a variety of existing datasets: LAION [107], ImageNet [23], COCO [74], FFHQ [59], CelebA [76], MetFaces [60], AFHQ [13], Forchheim [44], IMD2020 [86], Landscapes HQ [112], and VISION [111].³

3.4. Curating the evaluation set

We construct our evaluation set using the incompatible models from our automated sampling pipeline, commercial models (Sec. 3.2), and manually collected open source models. The evaluation set comprises 26K images sampled from 21 models not included in the training set.

³Following common convention, we refer to these images as *real* images, even though they may be synthetic (e.g., graphic design). More precisely, our goal is to distinguish “AI-generated” images from the originals.

This includes our commercial models set and an additional 11K images from 10 models: Deci Diffusion V2 [120], GALIP [119], KandinskyV2.2 [110], Kvikontent [66], LCM-LoRA-SDv1.5, LCM-LoRA-SDXL, LCM-LoRA-SSD1B [79], Stable Cascade [93], DF-GAN [118], and HDiT [16], sampled using RAISE [19], ImageNet [23], FFHQ [59], and COCO [74]-based captions.

The generated images are paired with the source real data that are used to prompt the generators. However, since some of the real datasets do not have appropriate licenses for redistribution (e.g., LAION [107, 108]), we created a *public* version of our evaluation set by pairing the generated images with openly licensed COCO [74] and FFHQ [59] which allow redistribution for non-commercial purposes. The *public* version of our evaluation set will serve as an easily reproducible and shareable evaluation set that will complement our default set. We will refer to our default set as the *comprehensive* evaluation set. We will also release the instructions to reconstruct our *comprehensive* set. However, note that it may not be possible to exactly reconstruct this set in the future due to link rot.

3.5. Generating images

Unconditional models are sampled until we reach the desired number of images. For class conditional models, we sample an equal number of images per class. Text-conditional models are sampled using captions obtained from real images (Sec. 3.3). We either use captions that are already present in the dataset or use BLIP [69] to generate them. Some models such as GigaGAN [57] and HDiT [16] do not provide a pretrained model, so we instead use their pre-generated images. Generated images are saved in PNG format to avoid compression artifacts. However, Firefly [1] generated images are saved in JPEG format as their web UI only allows downloading in JPEG.

4. Experiments

We use our dataset to conduct a study of generalization in visual forensics, asking a number of questions: (1) How well do forensics models trained on our dataset generalize to unseen models? (2) Does adding more models improve detection performance? (3) How does diversity of the training data affect performance? (4) What architectures and data augmentation schemes are most successful?

4.1. Training image forensics models

We train binary classifiers that detect generated images using our dataset to study the generalization in image forensics. We construct our training set of 5.4M images by pairing 2.7M generated images with 2.7M real images.

Training and evaluation setup. We evaluate the models trained on our dataset and compare them with prior works [43, 87, 125, 133]. As RED140 [43] already contains RED116 [5], we do not compare with RED116 dataset.

Following prior works [87, 125], we use the threshold-independent mean average precision (mAP) and accuracy (Acc.) as our evaluation metrics. We compute the mAP and accuracy by averaging the results of each generative model. We use five evaluation sets: Wang *et al.* [125], Ojha *et al.* [87], Synthbuster [7], GenImage [133], and our evaluation set. All evaluation sets apart from GenImage [133] evaluate out-of-distribution performance for all classifiers. GenImage [133] evaluation set, however, contains the same set of generators used in training, and is an in-distribution evaluation set for their classifiers. Concretely, the evaluation set by Wang *et al.* [125] and Ojha *et al.* [87, 125] contains models such as CRN [10], CycleGAN [132], DALL-E [88], DeepFake [25], IMLE [70], SAN [18], StarGAN [12], and SITD [9] which are unseen by both their and our classifiers. Synthbuster [7] evaluation set is comprised of RAISE [19]-based synthetic images of DALL-E [88, 89], Firefly [1], Glide [85], Midjourney [81], and Stable Diffusion [94, 100], and is mostly out of distribution for all classifiers. GenImage [133] evaluation set is a validation split of their training set; they use an identical set of models to train their classifier: ADM [24], BigGAN [8], Glide [85], Midjourney [81], Stable Diffusion [100], VQ-Diffusion [41], and Wukong [127].

Model architecture. Building on prior works which mainly used CLIP-ViT [27, 54, 96] and ResNet-50 [45], we consider ViT [27] and ConvNeXt [77] pretrained models for our classifiers. We use a plain ViT-S backbone [27] pretrained on CLIP objective [54, 96] using LAION-2B [108], ImageNet 21K, and ImageNet 1K datasets [23]. We also experiment with a ConvNeXt-S model [77] pretrained on ImageNet 21K and ImageNet 1K datasets [23]. We replace the classification head with a linear layer with sigmoid activation that outputs the probability of the image being generated. Unlike prior works [15, 87] that freeze the CLIP-ViT backbone, we train the backbone end-to-end. The models are obtained through `timm` [11, 126] library on Hugging Face. We experiment with two input resolutions, 224^2 and 384^2 , to evaluate the impact of the input resolution on the detector’s performance. We denote the detector with 384^2 input resolution as *High res*. We implement the models using PyTorch [91]. The hyperparameters are detailed in the supplement.

Data augmentation. Prior work considered augmentations that were designed to simulate postprocessing, such as flipping, cropping, Gaussian blur, and JPEG recompression to train their detectors [7, 15, 87, 125]. We propose an augmentation scheme that extends this approach and compare it with previously proposed augmentation methods. We expand the set of augmentations to handle additional transformations that can occur in the wild, such as padding, resizing, rotation, and shear, and integrate them into a framework that can apply complex sequences of transformations. We introduce a modified version of RandAugment [17] that applies a

Model	Evaluation Set (mAP)							Evaluation Set (Acc)						
	Wang et al. [125]		Ojha et al. [87]	SB [7]	GenImage [133]	Ours		Wang et al. [125]		Ojha et al. [87]	SB [7]	GenImage [133]	Ours	
						Comp.	Public						Comp.	Public
Wang et al. [125]	0.897	0.696	0.516	0.642	0.537	0.600	0.648	0.714	0.527	0.508	0.533	0.513	0.517	0.552
Ojha et al. [87]	0.939	0.957	0.620	0.797	0.592	0.656	0.760	0.791	0.821	0.532	0.641	0.540	0.548	0.646
GenImage [133]	0.929	0.984	0.813	0.999	0.912	0.968	0.934	0.795	0.966	0.719	0.990	0.818	0.886	0.862
♦RED140 [43] - High res.	0.900	0.954	0.765	0.927	0.764	0.861	0.862	0.694	0.780	0.558	0.674	0.562	0.565	0.639
Ours	0.964	0.991	0.904	0.990	0.971	0.977	0.966	0.873	0.950	0.818	0.946	0.861	0.888	0.889
Ours - High res.	0.967	0.996	0.974	0.998	0.987	0.994	0.986	0.901	0.970	0.908	0.957	0.892	0.912	0.923

Table 2. **Generalization of AI-generated image detectors across benchmarks.** We evaluate the classifiers trained on our dataset on several benchmarks, including our own. We also evaluate several previously released classifiers. Our *Comprehensive* set (abbreviated as *Comp.*) pairs the generated images with original real data; the *Public* set pairs them with openly licensed COCO [74] and FFHQ [59] for license-compliant redistribution of the evaluation set (Sec. 3.4). We use plain CLIP-ViT-S [27, 54, 96] architecture with 224^2 and 384^2 (*High res.*) input resolutions, Wang et al. [125] and GenImage [133] use ResNet-50 [45] with 224^2 input resolution, and Ojha et al. [87] uses CLIP-ViT-L with 224^2 input resolution as the backbone. Our classifiers show robust performance across all evaluation sets, outperforming all baselines in out-of-distribution settings ([7, 87, 125] and *Ours*) and nearly matches GenImage [133] on its *in-distribution* evaluation set. ♦: RED140 [43] is trained following our training procedure with 384^2 input resolution as they do not provide a trained classifier.

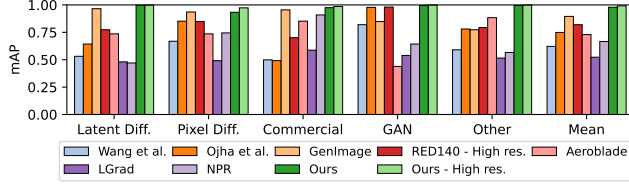


Figure 3. **Performance across generator types.** We evaluate the classifier performance across five generator types – latent diffusion, pixel diffusion, commercial models, GANs, and other architecture (Stable Cascade [93]), and report the mean. We additionally evaluate three forensics methods, Aeroblade [99], LGrad [116], and NPR [117]. Our classifiers show robust performance across all generator types, whereas prior works struggle to generalize.

randomly-ordered sequence of augmentations to the images. Specifically, our modified RandAugment samples a random number n between 0 and $n_{\max}$ for each augmentation type. Then, it applies the augmentations in random order until n augmentations are applied for each augmentation type to the image. We use various augmentations, including in-memory JPEG compression, random resizing with random interpolation methods, cropping, flipping, rotation, translation, shear, padding, and cutout.

4.2. Generalization across benchmarks

We first evaluate how well classifiers trained on our dataset perform across benchmarks. In Table 2, we observe that our models outperform the prior works [43, 87, 125, 133] on all evaluation sets except GenImage. This is expected since the GenImage evaluation set is a validation split of their training set; all of the generators are already seen by their classifier. On all other *unseen* benchmarks, our classifiers outperform all prior works. Notably, our classifiers achieve very high performance (0.987 mAP and 89.2% accuracy) on our out-of-distribution, *comprehensive* evaluation set, with a significant margin over prior works. This gap in performance can be traced to our training data which in-

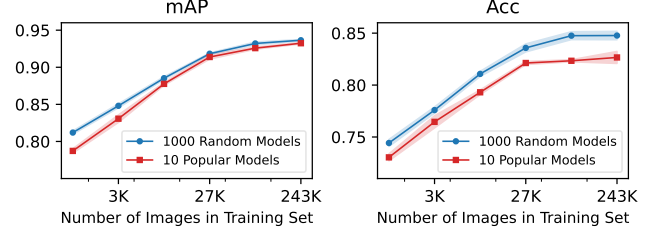


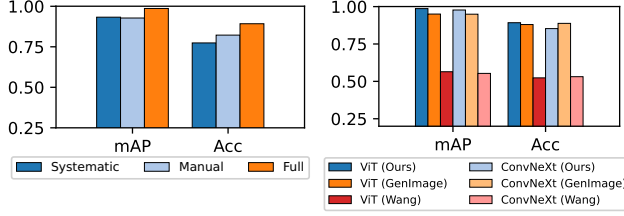
Figure 4. **Performance with increasing number of images.** We train a classifier with varying numbers of images from two sets: 1000 randomly chosen models and 10 popular (highly downloaded) models in the systematically collected subset. We report the mean and standard error bands for each data point across 4 randomly sampled subsets. The classifier trained from 1000 random models outperforms 10 popular models in all cases. Notably, the accuracy gap is wider than that of mAP, which may suggest that having a diverse set of models improves accuracy threshold calibration.

corporates a substantially richer variety of generators than existing works. Consequently, our classifiers demonstrate robust generalization to out-of-distribution data, where prior works often struggle.

To better illustrate the generalization of the classifiers, in Figure 3, we split our *comprehensive* evaluation set into five subsets and evaluate them: latent diffusion, pixel diffusion, commercial models, GANs, and other architecture type (Stable Cascade [93]). Furthermore, we evaluate three additional forensics methods: Aeroblade [99], LGrad [116], and NPR [117]. Our classifiers show strong performance across all generator types, unlike prior works which struggle to generalize. For the following experiments, we use our best-performing model (*High res.*) unless stated otherwise.

4.3. Impact of model diversity

Next, we examine the impact of the number of models in training data. We train classifiers with images sampled from 3 to 3333 generators and evaluate them (Fig. 1). To ensure that the gains are not due to simply sampling qualita-



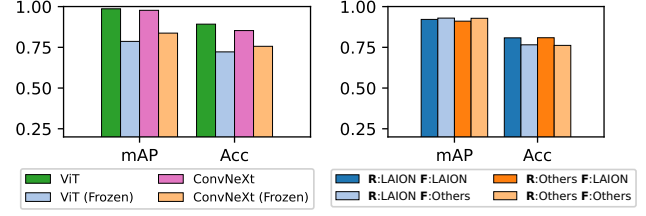
(a) Impact of generator type diversity (b) Classifier backbone comparison

Figure 5. (a) **Performance and model diversity.** We compare detection performance for commercial models using classifiers trained on different subsets of the dataset: the systematically collected latent diffusion models, the manually chosen models containing diverse generator types, and both. As diversity increases, so does performance. (b) **Classifier backbone comparison.** We compare the architectures across datasets: ours, GenImage [133], and Wang *et al.* [125]. Performance is similar between architectures.

tively different architectures, we only use our systematically collected latent diffusion models. For each data point, we sample 10 random subsets of models with 100K training images each and report the mean and standard error bands. We use an extended evaluation set that includes non-latent diffusion generators from our training set, which allows us to comprehensively assess the generalization capability of the classifiers trained exclusively on latent diffusion models. We find that the performance steadily increases with the number of models. However, the performance begins to flatten out beyond 1000 models, suggesting diminishing returns. Interestingly, the performance also improves on out-of-distribution architectures such as GANs and pixel-based diffusion models, even though the classifier is only trained on latent diffusion models.

In Figure 4, we vary the number of images from two sets: 1000 randomly chosen models and 10 popular models (as denoted by their number of downloads) downloaded from our systematically collected diffusion models. While the results show that the performance improves with more training images, it begins to plateau at approximately 27K images. Moreover, the classifier trained on 1000 models outperforms the 10 models in all cases, indicating that model diversity is important for strong performance. We also note that the accuracy gap is noticeably wider than that of mAP, which may suggest that model diversity is crucial in calibrating the accuracy thresholds of the classifiers.

Our experiments show that the performance improvements from increasing the number of models may plateau when they are limited to a single generator type (Fig. 1). In Figure 5a, we show that the diversity of the generator type also plays a major role in generalization. We train classifiers on three different sets of training data: our systematically collected set, manually chosen set, and a full set consisting of both subsets. The *systematic* set comprises entirely of latent diffusion models, and the *manual* set contains numerous generator types, including GANs, latent and pixel-based dif-



(a) Impact of frozen backbones (b) Semantic alignment analysis

Figure 6. (a) **Evaluating frozen backbones.** Freezing the pre-trained backbone, a common practice in prior works [15, 87], consistently decreases the performance. (b) **Analyzing source and generated data alignment.** We evaluate how the pairing of the real datasets affects performance. ‘R’ denotes the real dataset used in training, and ‘F’ indicates the source dataset used to obtain the captions for prompting the generators. The results suggest that pairing the source data (i.e., real data used to prompt the generators) with the generated images is not essential for performance.

fusion, and autoregressive models (Sec. 3.2). The classifier trained on the *manual* set with more diverse generator types shows similar performance compared to the one trained on the *systematic* set, despite containing fewer than half the number of images (774K vs. 1.9M). Additionally, we find that the two sets are complementary; the performance is further improved when we train using both sets.

4.4. Analysis of design choices

We examine the impact of various design choices, including some suggested in earlier works. In particular, we investigate the choice of backbone models, freezing the backbone, semantic alignment between the real and generated data, and robustness to transformations.

Classifier backbone comparison. We compare the performance of the classifier trained using CLIP-ViT [27, 54, 96] and ConvNeXt [77] backbones following our training procedure in Figure 5b. We examine three datasets: ours, GenImage [133], and Wang *et al.* [125]. We observe similar performance between architectures across all datasets.

Frozen backbone. Prior works [15, 87] suggested using a frozen CLIP-ViT backbone for training the classifiers. We study this by training the classifiers with both frozen and unfrozen pretrained backbones, using CLIP-ViT [27, 54, 96] and ConvNeXt [77]. As shown in Fig. 6a, freezing the backbone consistently leads to worse results, indicating that end-to-end training is crucial for high performance.

Semantic alignment. Existing works often pair the generated images with the source dataset (i.e., the real dataset used to prompt or generate the images) arguing that misaligned data can introduce bias [7, 15, 87, 125]. We test this practice in Fig. 6b by examining the performance with both semantically aligned and misaligned real datasets. Specifically, we consider two real datasets: one comprised exclusively from LAION [107] and another combining ImageNet [23],

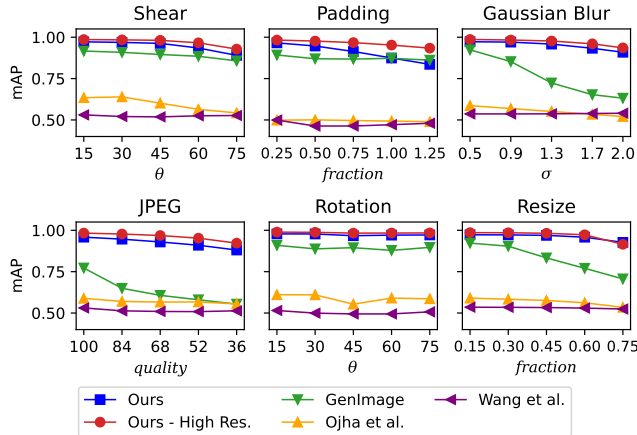


Figure 7. **Robustness to various transformations.** Our classifiers display robust performance across transformations. Other works generally show more sensitivity to these factors.

MS-COCO [74], LandscapesHQ [112], Forchheim [44], VISION [111], and IMD2020 [86]. We sample our systematically collected latent diffusion models using these two sets and categorize the generated images by their source dataset. The performance differences of all pairs were marginal, suggesting that strict alignment may not be as critical with sufficiently large data.

Robustness to transformations. Figure 7 illustrates the robustness of the classifiers against various transformations. Following prior works [87, 125], we test robustness to JPEG compression and Gaussian blur. Additionally, we examine robustness to rotation, resizing, padding, and shear, as they commonly occur in real-world scenarios. For *padding*, we randomly pad the width or height of the image with a given fraction and scale it back to the original size. Similarly, in *resize*, we randomly upsample or downsample the height and width of the image by a given fraction and resize it back to the original size (e.g., if *fraction* is 0.3, resize the height and width to $0.7\times$ or $1.3\times$ and then scale it back to the original size). The results demonstrate that our models are more robust to transformations than existing models. Specifically, GenImage [133] is notably more sensitive to Gaussian blur, JPEG compression, and resizing artifacts; classifier by Ojha *et al.* [87] displays sensitivity to *shear* transforms, and the one by Wang *et al.* [125] performs poorly overall.

4.5. Other forensics applications

Our dataset may enable further forensics studies that can take advantage of our diverse array of generators. To illustrate this, in the supplement, we identify the type of generator used to synthesize a given image by using k -nearest neighbor in the feature space of our classifier.

5. Conclusion

In this paper, we studied the problem of generalizing to unseen generative models in synthetic image detection. We proposed a new dataset, *Community Forensics*, which contains 4803 models and 2.7M images collected from various public sources. We studied the impact of model diversity and demonstrated that it plays a crucial role in enhancing data diversity and generalization performance. We trained classifiers on our dataset, studied their ability to generalize in various settings, and evaluated previously proposed models and training practices.

We do not intend for our dataset to be used to train classifiers that are directly used in the wild. Detecting in-the-wild synthetic images remains a challenging open problem, and detection errors can have severe consequences (e.g., falsely accusing an author of creating fake images or allowing misinformation to be certified as real). We hope that our work will serve as a stepping stone for future research in this area by providing tools and insights for studying generalization and data collection strategies.

Limitations. While our dataset is diverse, a large portion of the data is diffusion-based, especially models based on Stable Diffusion [100]. Despite this, Figure 1 demonstrates that adding more diffusion models to the training data improves the generalization capability of the classifiers across other generator types. Furthermore, Figure 5a illustrates improved classifier performance when combining the diffusion-only *systematic* set with a more diverse *manual* set. These results suggest that despite their similarities, each diffusion model may uniquely contribute through its semantic content, image processing pipelines, or architectural variations (Sec. 3.1). Future work may explore collecting more diverse models, including GANs, VQ-VAEs [122], and autoregressive models. We also note that the generative models sourced from the community may contain inappropriate content. While in many contexts it is important to detect such images, these generated images may require further scrutiny before being used in other downstream applications. Finally, although our experiments suggest that our forensics classifiers generalize to unseen models better than those of previous work, their error rates are still too high for many critical applications.

Acknowledgements. We thank the creators of the many open source models that we used to collect the Community Forensics dataset. We thank Chenhao Zheng, Cameron Johnson, Matthias Kirchner, Daniel Geng, Ziyang Chen, Ayush Shrivastava, Yiming Dou, Chao Feng, Xuanchen Lu, Zihao Wei, Zixuan Pan, Inbum Park, Rohit Banerjee, and Ang Cao for the valuable discussions and feedback. This research was developed with funding from the Defense Advanced Research Projects Agency (DARPA) under Contract No. HR001120C0123.

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